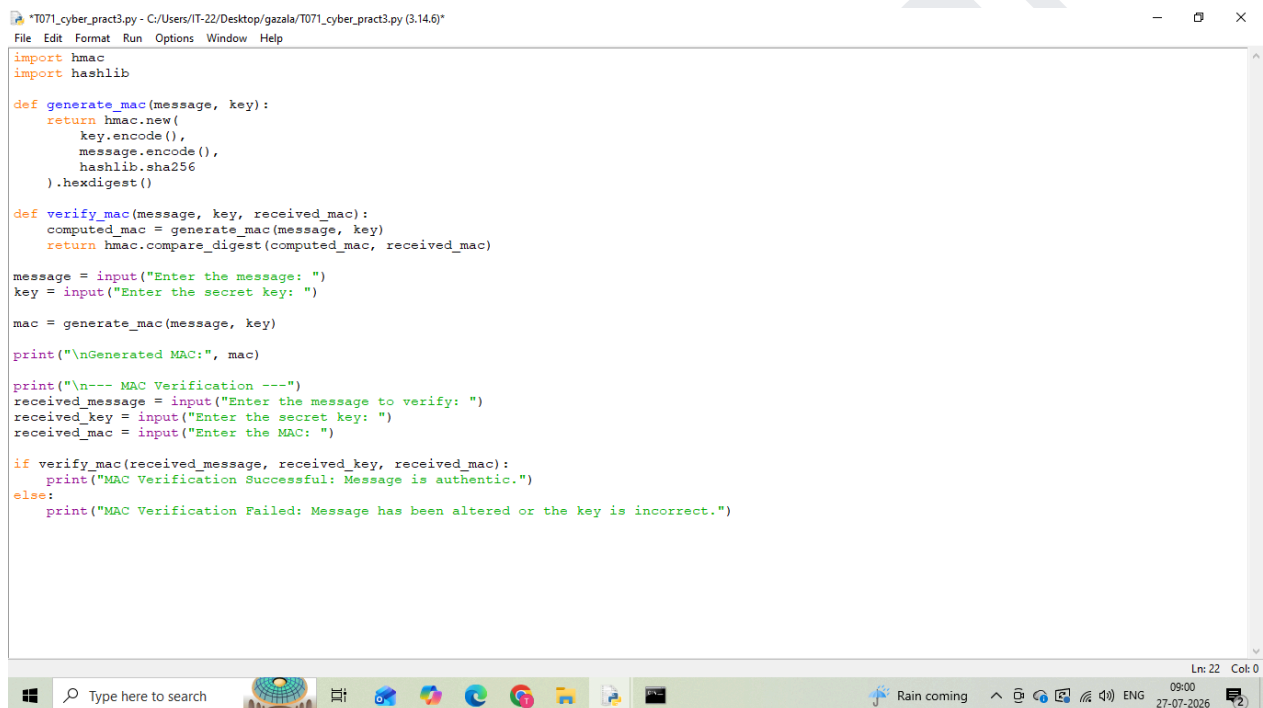


SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- CYBER SECURITY

Practical no: 3

Aim: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.



```
*T071_cyber_pract3.py - C:/Users/IT-22/Desktop/gazala/T071_cyber_pract3.py (3.14.6)
File Edit Format Run Options Window Help

import hmac
import hashlib

def generate_mac(message, key):
    return hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()

def verify_mac(message, key, received_mac):
    computed_mac = generate_mac(message, key)
    return hmac.compare_digest(computed_mac, received_mac)

message = input("Enter the message: ")
key = input("Enter the secret key: ")

mac = generate_mac(message, key)

print("\nGenerated MAC:", mac)

print("\n--- MAC Verification ---")
received_message = input("Enter the message to verify: ")
received_key = input("Enter the secret key: ")
received_mac = input("Enter the MAC: ")

if verify_mac(received_message, received_key, received_mac):
    print("MAC Verification Successful: Message is authentic.")
else:
    print("MAC Verification Failed: Message has been altered or the key is incorrect.")
```

Ln: 22 Col: 0

Windows taskbar: Type here to search, Rain coming, 09:00, 27-07-2026

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SUBJECT:- CYBER SECURITY

Code:

```
import hmac
import hashlib

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message = input("Enter the message: ")
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mac = generate_mac(message, key)

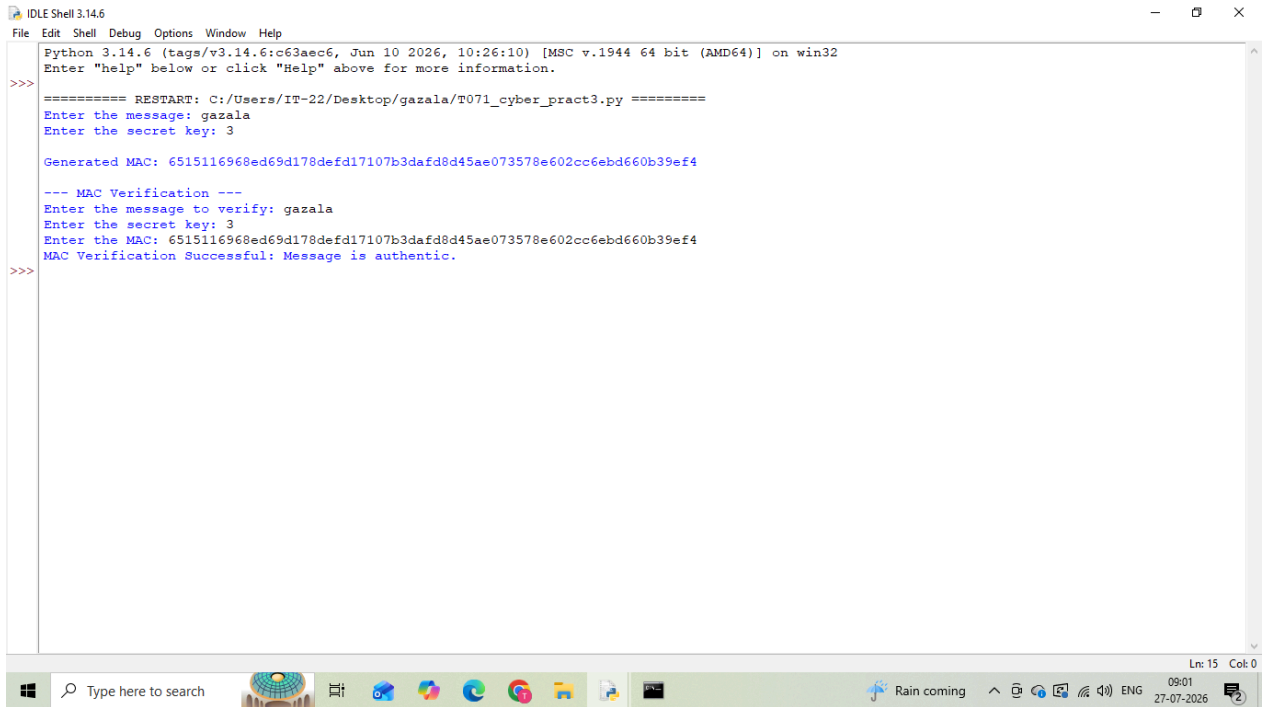
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```

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SUBJECT:- CYBER SECURITY



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-22/Desktop/gazala/T071_cyber_pract3.py =====
Enter the message: gazala
Enter the secret key: 3

Generated MAC: 6515116968ed69d178defd17107b3dafd8d45ae073578e602cc6ebd660b39ef4

--- MAC Verification ---
Enter the message to verify: gazala
Enter the secret key: 3
Enter the MAC: 6515116968ed69d178defd17107b3dafd8d45ae073578e602cc6ebd660b39ef4
MAC Verification Successful: Message is authentic.
>>>
```

Ln: 15 Col: 0

Type here to search

Rain coming

09:01
27-07-2026